

ATTENDANCE MANAGEMENT DASHBOARD

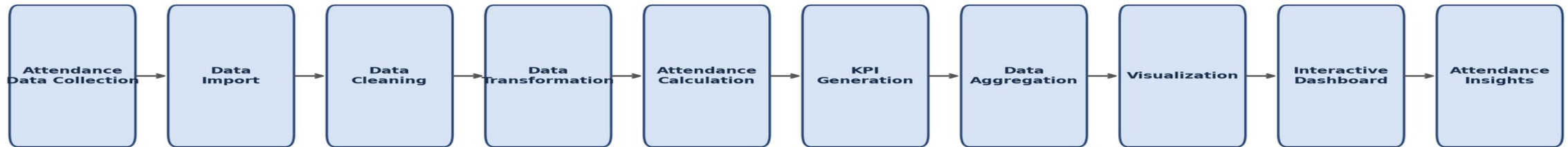
Project Summary — Algorithm, Dataset, Implementation & Results

Objective

- 1 Automate calculation of student attendance percentages from raw records
- 2 Generate KPIs — total students, average attendance, present/absent counts, low-attendance count
- 3 Summarize attendance by student, subject, class/section, and time period
- 4 Present insights through interactive charts, tables, and KPI cards
- 5 Enable filtering by student, subject, class/section, and date range
- 6 Flag students below an attendance threshold for timely intervention

Methodology

Descriptive analytics pipeline — monitoring & visualization, not predictive ML



Collect & Import

Gather attendance records; import from Excel/CSV

Clean & Standardize

Fix missing values, duplicates, inconsistent text

Calculate & Aggregate

Attendance %, KPIs, multi-level summaries

Visualize & Validate

Charts, KPI cards, interactive filters, checks

Dataset

Column	Description	Type
Student_ID	Unique identifier of the student	Text / Integer
Student_Name	Name of the student	Text
Class_Section	Class or section	Text
Subject	Subject name or code	Text
Attendance_Date	Date of attendance	Date
Attendance_Status	Present or Absent	Categorical
Faculty	Faculty associated with the subject	Text

Demo Dataset Stats

4,405

Raw records generated

29

Missing status values found

5

Duplicate records found

4,371

Records after cleaning

Algorithm

$$\text{Attendance Percentage} = (\text{Classes Attended} / \text{Total Classes}) \times 100$$

1. Collect attendance records
2. Import from Excel/CSV
3. Inspect for missing/duplicate/inconsistent data
4. Clean invalid or missing records
5. Standardize ID, section, subject, date, status
6. Transform into analysis-ready structure
7. Calculate present/absent counts & attendance %
8. Generate KPIs
9. Summarize by student, subject, section, period
10. Build charts, tables, KPI cards
11. Apply interactive filters
- 12–13. Present dashboard & validate results

Results

40

Total Students

85.61%

Average Attendance

3,742

Present Count

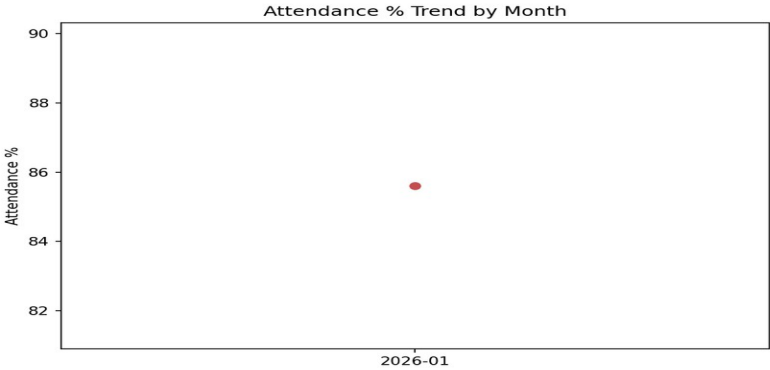
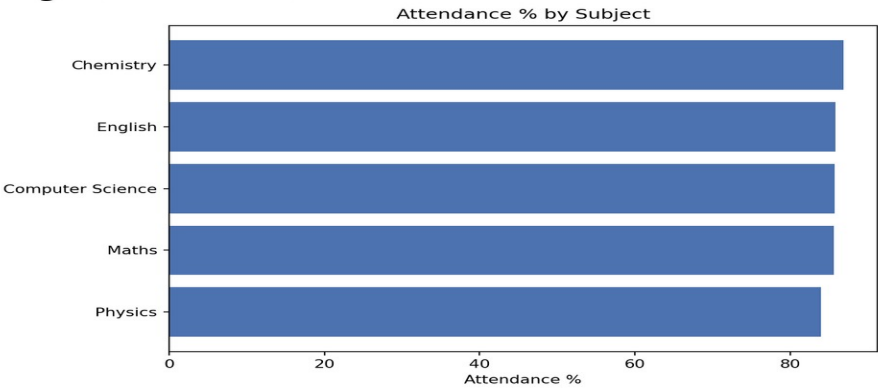
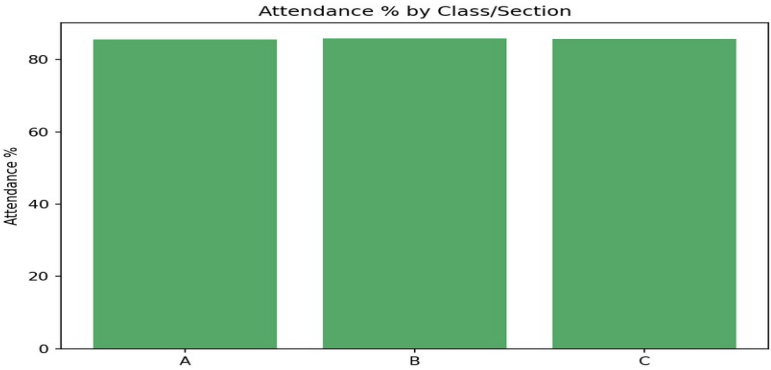
629

Absent Count

Attendance Management Dashboard

KPI SUMMARY

Total Students: 40
Total Classes Recorded: 4371
Average Attendance Percentage: 85.61
Total Present Count: 3742
Total Absent Count: 629
Low Attendance Student Count: 0



Validation

- ✓ Present count matches
- ✓ Total classes match
- ✓ All % values within 0-100 range

Conclusion

The Attendance Management Dashboard successfully converts raw attendance records into meaningful, interactive insights — reliably cleaning messy real-world data, computing accurate attendance percentages, and surfacing KPIs and visuals that let faculty quickly identify trends and students needing attention.

Future Scope

A future version could add a predictive ML component (e.g., attendance-risk forecasting) using the same cleaned dataset — at which point a train/test split would become relevant.

Thank You